

Golden Section Search — Algorithm, Code & Output

Algorithm

Finds minimizer of unimodal f on $[a,b]$ using one fixed ratio $\tau=(\sqrt{5}-1)/2\approx 0.618$ instead of bisection — self-similar, geometric shrink, no precomputed step count.

- 1) Place $x_1=b-\tau(b-a)$, $x_2=a+\tau(b-a)$; evaluate f at both.
- 2) Compare: $f(x_1)>f(x_2) \rightarrow$ drop left, $a \leftarrow x_1$; else drop right, $b \leftarrow x_2$. Reuse old point + its f -value as one new pair member $\rightarrow$ only 1 new eval/iteration ($\tau^2=1-\tau$ keeps it self-similar).
- 3) Repeat while $(b-a) > \text{tol}$; return midpoint $(a+b)/2$ as $x_{\min}$.

Code (short)

```
f <- function(x) (x-2)^2 * (x-8)^2

golden_search <- function(f,a,b,tol=1e-4){
  tau <- (sqrt(5)-1)/2
  hist <- data.frame(iter=0,a=a,b=b)
  x1<-b-tau*(b-a); x2<-a+tau*(b-a)
  f1<-f(x1); f2<-f(x2); iter<-0
  while((b-a) > tol){
    iter<-iter+1
    if(f1>f2){ a<-x1;x1<-x2;f1<-f2
      x2<-a+tau*(b-a); f2<-f(x2) }
    else{ b<-x2;x2<-x1;f2<-f1
      x1<-b-tau*(b-a); f1<-f(x1) }
    hist<-rbind(hist,data.frame(iter,a,b))
  }
  list(xmin=(a+b)/2, interval=c(a,b), history=hist)
}
result <- golden_search(f, a=1, b=5, tol=1e-4)
```

Output

Run on $f=(x-2)^2(x-8)^2$, $[1,5]$, $\text{tol}=1e-4 \rightarrow 23$ iterations.

$x_{\min} \approx 1.999998$ $\text{interval} \approx [1.999967, 2.000029]$

Bracket collapses onto true root $x=2$ (where $f=0$); width shrinks geometrically by τ each step. Plot: red/green lines (a,b per step) funnel tightly around $x=2$ over the f curve, visually showing the shrink.